

Modular Exponentiation

Module I

Modular arithmetic properties

Modular arithmetic exhibits the following properties.

1. $(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$
2. $(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$
3. $(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$

Methods to compute modular exponentiation

1. Brute force method
2. Factoring
3. Memory efficient
4. Successive squaring (power of 2)
5. General fast method

Brute Force method for modular exponentiation.

- This is a naïve method.
- Example: $3^4 \bmod 5 = 81 \bmod 5 = 1$
- As the exponent gets larger, it becomes harder. It takes longer time and quickly run out of memory.
- E.g. $3^{2096} \bmod 5$

factoring

- If an Integer e factors into $e=xyz$, then $a^e = a^{xyz} = (((a^x))^y)^z$
- Example: $14^{105} \bmod 5 = (((14^3))^5)^7 \bmod 5$ since $3 * 5 * 7 = 105$
- $14^3 = 2744 \bmod 5 = 4 \bmod 5 = 4$
- $4^5 = 1024 \bmod 5 = 4 \bmod 5 = 4$
- $4^7 = 16384 \bmod 5 = 4 \bmod 5 = 4$

Memory efficient

- Multiply by the base and reduce each iteration.
- Example: $3^7 \bmod 5$
- $3^2 = 3 * 3 = 9 \bmod 5 = 4$
- $3^3 = 3^2 * 3 \bmod 5 = 4 * 3 \bmod 5 = 12 \bmod 5 = 2$
- $3^4 = 3^3 * 3 \bmod 5 = 2 * 3 \bmod 5 = 6 \bmod 5 = 1$
- $3^5 = 3^4 * 3 \bmod 5 = 1 * 3 \bmod 5 = 3 \bmod 5 = 3$
- $3^6 = 3^5 * 3 \bmod 5 = 3 * 3 \bmod 5 = 9 \bmod 5 = 4$
- $3^7 = 3^6 * 3 \bmod 5 = 4 * 3 \bmod 5 = 12 \bmod 5 = 2$

Successive squaring

- Useful if the exponent is power of 2
- Example: $3^{128} \bmod 7$
- $3^2 \bmod 7 = 3 * 3 = 9 \bmod 7 = 2$
- $3^4 \bmod 7 = 3^2 * 3^2 \bmod 7 = 2 * 2 \bmod 7 = 4 \bmod 7 = 4$
- $3^8 \bmod 7 = 3^4 * 3^4 \bmod 7 = 4 * 4 \bmod 7 = 16 \bmod 7 = 2$
- $3^{16} \bmod 7 = 3^8 * 3^8 \bmod 7 = 2 * 2 \bmod 7 = 4 \bmod 7 = 4$
- $3^{32} \bmod 7 = 3^{16} * 3^{16} \bmod 7 = 4 * 4 \bmod 7 = 16 \bmod 7 = 2$
- $3^{64} \bmod 7 = 3^{32} * 3^{32} \bmod 7 = 2 * 2 \bmod 7 = 4 \bmod 7 = 4$
- $3^{128} \bmod 7 = 3^{64} * 3^{64} \bmod 7 = 4 * 4 \bmod 7 = 16 \bmod 7 = 2$

Fast exponentiation

- Find the binary expansion of the exponent.
- Write the exponent as the sum of its binary terms.
- Express the original exponentiation in terms of this sum.
- Compute the factors reduced by the modulus.
- Multiply together (Reducing whenever possible)

example 1: Find $3^{133} \bmod 7$

$$133_{10} = 10000101_2 = 128 + 4 + 1$$

$$3^{133} = 3^{128+4+1} = 3^{128} * 3^4 * 3^1 \bmod 7$$

From the successive squaring, $3^{128} \bmod 7 = 2$ and $3^4 \bmod 7 = 4$

$$3^{133} = 3^{128+4+1} = 3^{128} * 3^4 * 3^1 \bmod 7 = 2 * 4 * 3 \bmod 7 = 24 \bmod 7 = 3.$$

Modular exponentiation examples

- How to compute?

$$c = a^m \bmod n$$

example 1: $3^{10} \bmod 5$

$$3^2 \bmod 5 \Rightarrow 9 \bmod 5 \Rightarrow 4.$$

$$3^4 \bmod 5 \Rightarrow 3^2 * 3^2 \bmod 5 \Rightarrow (3^2 \bmod 5 * 3^2 \bmod 5) \bmod 5.$$

$$\Rightarrow (4 * 4) \bmod 5. \Rightarrow 16 \bmod 5 \Rightarrow 1$$

$$3^8 \bmod 5 = 3^4 * 3^4 \bmod 5 = (3^4 \bmod 5) * (3^4 \bmod 5) \bmod 5 = 1 \bmod 5 = 1$$

$$3^{10} \bmod 5 = 3^8 * 3^2 \bmod 5 = (3^8 \bmod 5 * 3^2 \bmod 5) \bmod 5 = 1 * 4 \bmod 5 = 4$$

Modular exponentiation

example 2: $2^{34} \bmod 3$

$$2^2 \bmod 3 \Rightarrow 4 \bmod 3 \Rightarrow 1.$$

$$2^4 \bmod 3 \Rightarrow 2^2 * 2^2 \bmod 3 \Rightarrow (2^2 \bmod 3 * 2^2 \bmod 3) \bmod 3.$$

$$\Rightarrow (1 * 1) \bmod 3. \Rightarrow 1 \bmod 3 \Rightarrow 1$$

$$2^8 \bmod 3 = 2^4 * 2^4 \bmod 3 = (2^4 \bmod 3) * (2^4 \bmod 3) \bmod 3 = 1 * 1 \bmod 3 = 1$$

$$2^{16} \bmod 3 = 2^8 * 2^8 \bmod 3 = (2^8 \bmod 3 * 2^8 \bmod 3) \bmod 3 = 1 * 1 \bmod 3 = 1$$

$$2^{32} \bmod 3 = 2^{16} * 2^{16} \bmod 3 = (2^{16} \bmod 3 * 2^{16} \bmod 3) \bmod 3 = 1 * 1 \bmod 3 = 1$$

$$2^{34} \bmod 3 = 2^{32} * 2^2 \bmod 3 = (2^{32} \bmod 3 * 2^2 \bmod 3) \bmod 3 = 1 * 1 \bmod 3 = 1$$

Modular exponentiation-examples

$$2^{33} \bmod 30, 3^{100} \bmod 29, 23^3 \bmod 30$$

Example 3:

$$23^3 \bmod 30 = -7^3 \bmod 30 \parallel 23 \bmod 30 \text{ can be } 23 \text{ or } -7.$$

$$= -7^2 * 7^1 \bmod 30 = 49 * -7 \bmod 30 = -133 \bmod 30 = -13 \bmod 30 = 17 \bmod 30 = 17.$$

Example 4:

$$31^{500} \bmod 30 = 1^{500} \bmod 30 = 1 \bmod 30 = 1.$$

Example 5:

$$242^{329} \bmod 243 = -1^{329} \bmod 243 = -1 \bmod 243 = 242.$$

Example 6:

$$\begin{aligned} 11^7 \bmod 13 &= 11 \bmod 13 * 11 \bmod 13 * 11 \bmod 13 * 11 \bmod 13 * 11 \bmod 13 * 11 \bmod 13 * 11 \bmod 13 \\ &= -2 * -2 * -2 * -2 * -2 * -2 * -2 \bmod 13 \text{ (Iterative)} \\ &= -128 \bmod 13 \\ &= -11 \bmod 13 = 2 \end{aligned}$$

Modular exponentiation-examples

Example 7:

$$88^7 \bmod 187 =$$

$$= 88^1 \bmod 187 = 88$$

$$= 88^1 * 88^1 \bmod 187 = 88 * 88 \bmod 187 = 7744 \bmod 187 = 77$$

$$= 88^4 \bmod 187 = 88^2 * 88^2 \bmod 187 = 77 * 77 \bmod 187 = 5929 \bmod 187 = 132$$

$$= 88^7 \bmod 187 = 88^4 * 88^2 * 88 \bmod 187 = 132 * 77 * 88 \bmod 187 = 894432 \bmod 187$$

$$= 11$$

Modular exponentiation-examples

What is the last two digits of 29^5

Example 8:

$$29^5 \bmod 100 =$$

$$= 29^1 \bmod 100 = 29 \text{ or } -71$$

$$= 29^2 \bmod 100 = 841 \bmod 100 = 41 \bmod 100 = 41 \text{ or } -59$$

$$\begin{aligned} &= 29^5 \bmod 100 = 29^2 * 29^2 * 29 \bmod 100 = 41 * 41 * 29 \bmod 100 = 1681 * \\ &29 \bmod 100 = (1681 \bmod 100 * 29 \bmod 100) \bmod 100 = 81 * 29 \bmod 100 \end{aligned}$$

$$= 2349 \bmod 100 = 49$$